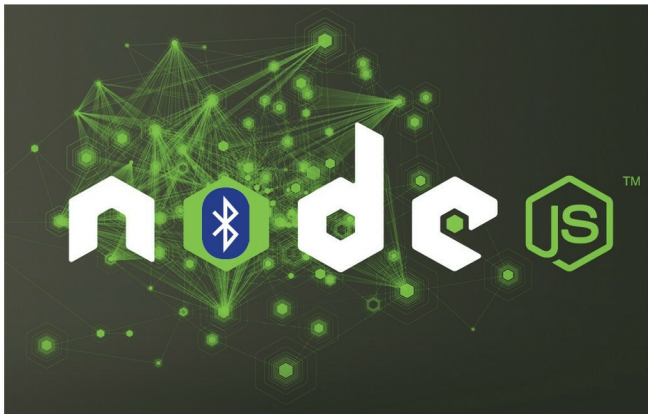


Talk to Your Bluetooth Device Using Node.js

Node.js is an event-driven I/O server-side JavaScript environment based on Google's Chrome V8. It is a powerful runtime environment for developing varied tools and applications. This article talks about interfacing Node.js with Bluetooth devices.



Change in the world of technology is constant. This requires every one of us to upgrade ourselves with new technologies and adapt to them. With the Internet of Things (IoT) emerging as a new technology, companies are building new ecosystems for 'things' so that they too can have a piece of cake. Open source frameworks facilitated by the Linux Foundation are adding a dimension to the open things ecosystem.

Embedded computing, which once used to be closer to hardware languages like C, is slowly moving away to high level languages like Java. The reasons for such movement could be multi-fold, ranging from high power hardware that can still provide a better user experience, to the scalability of new generation applications. Such movement necessitates newer frameworks for the high level languages to be able to adapt to the new ecosystem. Node.js is one such framework that is getting adopted to develop a wide range of solutions.

An overview of Node.js

Node.js is an open source JavaScript runtime framework built on Chrome's V8 JavaScript engine. The community activity is governed by the Node.js Foundation and the framework is available from <https://nodejs.org>. The Node.js framework has the largest ecosystem of open source libraries, which extend its functionality considerably. These packages are available at <https://www.npmjs.com/>. The implementation of the framework can be explored at <https://github.com/nodejs>. Having understood in brief how the framework works, let's explore it in detail and understand the internal architecture.

The Node.js framework is part of the server side JavaScript, with the goal of offering a scalable framework to develop high performance network applications. This goal is realised by providing a single-threaded, event driven, non-blocking I/O framework. The framework is implemented in C/C++ with interface APIs in JavaScript. The block diagram